Fiscal Year:	FY 2008	**Task Last Updated:**	FY 06/19/2008
PI Name:	Wang, Ya M.D., Ph.D.		
Project Title:	The mechanism of excess relative risk on carcinogenesis induced by high-LET radiation		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	30322	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2007 Space Radiation NNJ07ZSA001N
Start Date:	09/01/2007	**End Date:**	08/31/2008
No. of Post Docs:		**No. of PhD Degrees:**	1
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 8/31/2008, per J. Dardano/JSC. PI left Thomas Jefferson Medical at that time. Grant has since been set up for continuation at Emory University for FY2009 and beyond (jp, TB editor, 6/4/09)		
Key Personnel Changes/Previous PI:	No changes		
COI Name (Institution):	Wang, Hongyan (Thomas Jefferson University)		
Grant/Contract No.:	NNX07AT40G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Our hypothesis in this application is that high LET radiation induced excess relative risk on mutagenesis as well as on carcinogenesis is due to inefficient Ku-dependent non-homologous end-joining (NHEJ) repair. The results from this project will provide a mechanism to explain why high LET radiation induced such excess relative risk (ERR), and will significantly improve our understanding of the relationship between DNA repair and high LET radiation induced such ERR. Therefore, the results from this proposal will assist NASA Space Radiation Program to reach its major goal by providing specific parameters for evaluating high LET radiation induced ERR on mutagenesis as well as on carcinogenesis and by providing protective targets for reducing high LET radiation induced mutagenesis as well as carcinogenesis.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The study described in this application will determine whether the major reason for high linear energy transfer (LET) ionizing radiation (IR) to induce excess relative risk (ERR) as compared with low LET IR on carcinogenesis is because of inefficient Ku-dependent non-homologous end-joining repair. The results from this proposal will answer the question whether inefficient Ku-dependent repair is the major reason for high LET IR to induce ERR on carcinogenesis. Results from this proposal will assist the NASA Space Radiation Program to reach its major goal by providing specific parameters for evaluating the ERR on carcinogenesis and by providing specific protective targets for reducing high LET IR-induced ERR on the carcinogenesis.
Task Progress:	This project is funded since last year September. Since then, we have followed the proposal to carry out some experiments as described in our specific aims. This application is in response to the NASA Program NNJ07ZSA001N and addresses research related to reducing the risks of high LET radiation-induced cancer. High linear energy transfer (LET) ionizing radiation (IR), when compared with low LET IR, induces more death both in vitro as well as in vivo, induces more mutations in vitro and induces more cancers in vivo. IR-induced DNA double strand breaks (DSBs) are the severest threat for IR-induced killing as well as are the initial and the essential event for IR-induced genomic integrity. Two kinds of major DSB repair exist in eukaryotic cells, homologous recombination repair (HRR) and non-homologous end-joining (NHEJ). HRR is the major pathway for repairing DSBs in yeast, but both the NHEJ and HRR pathways are required for DSB repair in mammalian cells. NHEJ repair includes a Ku-dependent main pathway and a PARP-1 dependent complementary pathway. High LET IR induces more cell killing when compared with low LET IR and is described as relative biologic effectiveness (RBE), which is believed to be caused by ineffective rejoining of DNA DSBs involved in the Ku-dependent NHEJ pathway. However, it remains unclear whether high LET IR also affects the pathways of HRR and PARP-1-dependent complementary NHEJ. Our recent data suggest that high LET IR affects only the Ku-dependent main NHEJ but not other repairs, including HRR and PARP-1-dependent complementary NHEJ. Based on our new data, we hypothesize that the major reason for high LET IR to induce excess relative risk on carcinogenesis is because of inefficient Ku-dependent non-homologous end-joining (NHEJ) repair. The purpose of this project is to test our hypothesis and three specific aims are proposed to accomplish this goal. A.1. Determine whether the major reason for high LET IR inducing excess deaths in cells and in mice is because of inefficient Ku-dependent NHEJ repair. A.2. Determine whether the major reason for high LET IR inducing excess relative risk (ERR) on carcinogenesis in mice is because of inefficient Ku-dependent NHEJ repair. A.3. Determine whether the major reason for high LET IR inducing excess relative risk (ERR) on genomic instability is because of inefficient Ku-dependent NHEJ repair. We will use repair deficient mice combining in vitro and in vivo approaches to carry out our experiments. No matter if our hypothesis is right or not, results from these experiments will tell us how much a percentage of each repair pathway (Ku-dependent NHEJ, PARP-1-dependent NHEJ or HRR) is responsible for the high LET IR induced ERR on genomic instability, and will provide valuable information for predicting high LET IR induced carcinogenesis. During the funding period, by using the different cell lines that are deficient in different repair pathway, we demonstrated that only the Ku-dependent NHEJ pathway, and not the HRR pathway or the PARP-1-dependent NHEJ pathway, is inhibited by high LET IR. We suggest that this might be a consequence of more small fragments of dsDNA induced by high LET IR compared with low LET IR. We believe that these results will provide guidance for space radiation protection. These results are reflected in our manuscript, entitled “The Ku-dependent non-homologous end-joining but not other repair pathway is inhibited by high-linear energy transfer ionizing radiation.” that is published in the journal of DNA Repair, this year (listed in Bibliography section below). [NOTE from Task Book Editor, 6/4/2009: End date changed to 8/31/2008, per J. Dardano/JSC. PI left Thomas Jefferson Medical at that time. Grant has since been set up for continuation at Emory University for FY2009 and beyond --see PI Yang with title, "The mechanism of excess relative risk on carcinogenesis induced by high-LET radiation--NNX09AF24G."]
Bibliography Type:	Description: (Last Updated: 07/07/2021)
Articles in Peer-reviewed Journals	Wang H, Wang X, Zhang P, Wang Y. "The Ku-dependent non-homologous end-joining but not other repair pathway is inhibited by high-linear energy transfer ionizing radiation." DNA Repair (Amst). 2008 May 3;7(5):725-33. Epub 2008 Mar 5. PMID: 18325854 , May-2008
Articles in Peer-reviewed Journals	Wang H, Wang Y. "Checkpoint response plays a more protective role in HZE particle-irradiated cells than in X ray-irradiated cells." Cell Cycle. 2008 Aug;7(15):2444-5. Epub 2008 May 8. PMID: 18682684 , Aug-2008